

README: Data Extraction and Calculation for Ad Valorem Tariffs Rates and Forest Growing Stock and Forest Stock Growth for TiMBA

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Calculation of Ad Valorem Tariffs Rates

1. Overview

The Timber market Model for policy-Based Analysis **TiMBA** (TI-FSM 2024) is a partial economic equilibrium model for the global forest products market. It simulates production, consumption, and trade of wood-based products across 180 countries.

Trade in TiMBA depends, among others, on the transportation costs. Transportation costs entails commodity-specific freight costs and tariffs. Tariffs are calculated as a percentage of the value of the imported product using ad valorem rates. They are derived from the **World Trade Organization (WTO) Integrated Database (IDB)** (WTO 2024) notifications using **6-digit** (subheadings) or **4-digit codes** (headings) **HS (Harmonized System) codes**.

This Readme provides a detailed explanation of how the ad valorem rates were extracted and calculated for different wood-based products across different countries. The rates are based on the **Most-Favored-Nation (MFN)** applied tariff, and trade agreements were not considered in the calculations.

This research updates previous data as published with the Forest Sector Model GFPM (Buongiorno 2003; Buongiorno 2021). The base for the tariff update is the least available HS code version for each country, with updates based on data from **July 2024** sourced from the **World Trade Organization (WTO) Integrated Database (IDB)** (WTO 2024).

2. Notes on Tariff Data Calculation and Updates

I. HS Code Selection

The extraction and / or calculation of ad valorem rates is based on the **HS codes** for each product. The HS codes provide standardized classifications of goods, used internationally for tariff purposes. The HS classifies products using 6-digit codes that are organized by chapters (2 digits), headings (4 digits), and subheading (6 digits). If applicable, we extract information stated on the **4-digit level**. However, in some cases, a product in the TiMBA model framework is represented by **6-digit HS codes**.

II. Updates of Rates

The calculation uses the **least available HS version** for a particular country. The rates used in the TiMBA scenario input file were carefully checked for accuracy and updated in case they differ from the values indicated by Buongiorno (2021).

Differences between provided data and the information delivered with the GFPM (Buongiorno 2021) should come from the updates of the values. However, especially when combined HS code combinations are used for product representation, the difference may also arise from the different use of product classifications and assignments in the current work and the GFPM (Buongiorno 2003) (Buongiorno 2015)

III. Ad Valorem Rate Acquisition

In general, mean ad valorem rates for those specific products are extracted from the WTO IDB. Products represented by more than one HS-code are treated separately for calculating mean ad valorem rates for those specific products. Thus, depending on the number of HS codes digit levels involved, ad valorem rates were either extracted or calculated:

Where possible, the ad valorem rate for a specific country and product are extracted directly from the **WTO IDB** as (mean) ad valorem rates on the 4-digit or 6-digit code level. Prerequisite for the former case is, that the product representation in TiMBA is congruent with the 4-digit level in the HS system.

Some products are represented by more than one 4-digit and / or 6-digit HS code. When this occurs, the tariffs for these codes are treated separately for the calculation, and the final ad valorem rate reflects the aggregate or a representative value based on the available data. In case that a product in TiMBA is composed of **multiple HS codes (see Table 1)**, the **ad valorem rate is calculated** based on the available tariff data for the product and country as the average of ad valorem duties of the respective reporter country and product.

IV. List of Products

Table 1: Products and assignment of HS codes used for the ad valorem rate calculations.

Product	HS codes
Fuelwood	4401
Industrial Roundwood	4403
Sawnwood Non-coniferous	440721, 440722, 440723, 440724, 440725, 440726, 440727, 440728, 440729, 440791, 440792, 440793, 440794, 440795, 440796, 440797, 440799
Sawnwood Coniferous	440710, 440711, 440712, 440713, 440714, 440719
Veneer and Plywood	4412
Particleboard	4410
Fibreboard	4411
Mechanical Pulp	4701
Chemical and Semi-chemical pulp	4703, 4704, 4705
Other Fibre Pulp	4706
Waste Paper	4707
Newsprint	480100
PWP	4802
Paper and Paperboard	4804, 4805, 480620, 480630, 480640, 4807, 4808, 4810, 4811, 481500, 4819, 4821, 4823

Data Extraction and Calculation for Forest Growing Stock as well as Forest Stock Growth and Forest Area and Forest Area Growth

1. Overview

This dataset contains information on forest growing stock and forest stock growth as well as forest area and forest area growth for 180 countries, as considered in the TiMBA model. While data on forest area and forest area growth for the are adequately provided by the FAO Forest Resources Assessment (FRA), data on forest growing stock and forest stock growth are not available for all countries. This Manual explains the methodology used to extract data as well as fill in the data gaps and provide consistent values for forest growing stock and stock growth across all countries. The methodology ensures that the data used in the TiMBA model is as consistent and reliable as possible, despite the fragmented nature of global forest stock data.

2. Notes on Extraction and Calculation of Data

Whenever available, data for the country-specific growing stock for the total forest area was taken from the FRA 2020 for the reporting year 2020 and from FRA 2025 for TiMBA scenario input files with 2021 or later as starting year. This data provides the most up-to-date estimates for forest stock at a global level.

In cases where data for the target year was not reported but older data for forest growing stock data for the total forest area (e.g., from 2010) was available, we used them to fill in gaps where more recent data are unavailable. If no data on total forest area were reported at all, we searched the FAO FRA 2020 database for data specific to naturally regenerated and/or planted forest areas. In case of FAO FRA 2025 the latter case does not apply.

In some instances, total forest growing stock was calculated from the FRA biomass stock (measured in t/ha). This approach provides an alternative estimation based on biomass density when forest growing stock reports for any forest type were non-available.

For those countries not covered by the FRA reports, the total forest growing stock was calculated using data from the GlobalForestWatch on living woody biomass density. If the reported forest area was greater than 1 while the growing forest stock was reported and/or calculated to be smaller than 1, a correction was applied where the stock value was set to 1. This correction was necessary to standardize data across the dataset and ensure that unrealistic or inconsistent values were adjusted.

For the estimation of forest stock growth in TiMBA, the objective is to derive the biological growth of forest growing stock independent of harvest removals. Since the growing stock values reported in the FAO FRA represent observed stocks after harvest activities, these values were adjusted to approximate a hypothetical growing stock development without roundwood removals.

For TiMBA scenario input files with a starting year of 2020, annual forest stock growth rates were derived from the development of forest growing stock between 2015 and 2020. For scenario input files with starting years of 2021 or later, the respective period 2020–2025 was used.

To estimate the hypothetical growing stock development excluding harvest, annual roundwood supply data from FAOSTAT were aggregated over the corresponding reporting period and added back to the observed growing stock reported by FAO FRA. This adjustment reconstructs the growing stock level that would have been reached in the absence of harvest removals. For the period 2015–2020, the adjusted growing stock was calculated as:

$$GS_{2020}^* = GS_{2020} + \sum_{t=2015}^{2019} RW_t$$

For the period 2020–2025, the adjusted growing stock was calculated analogously:

$$GS_{2025}^* = GS_{2025} + \sum_{t=2020}^{2024} RW_t$$

where:

- GS_t^* = adjusted hypothetical growing stock in year T excluding harvest removals (million m³)
- GS_t = observed growing stock reported by FRA in year T (million m³)
- RW_t = annual roundwood supply in year t (million m³ over bark).

Based on these adjusted growing stock values, the average annual relative growth rate of forest growing stock excluding harvest was calculated using the compound annual growth rate (CAGR) over the respective five-year reporting period.

For the period 2015–2020:

$$g_{GS} = \left(\frac{GS_{2020}^*}{GS_{2015}} \right)^{\frac{1}{5}} - 1$$

For the period 2020–2025:

$$g_{GS} = \left(\frac{GS_{2025}^*}{GS_{2020}} \right)^{\frac{1}{5}} - 1$$

where:

- g_{GS} = average annual relative growth rate of forest growing stock excluding harvest.

This formulation assumes a constant compounded annual growth rate over the respective reporting interval and provides a harmonized estimate of biological forest stock growth independent of harvest removals.

If no sufficiently reliable data on growing stock development were available, the average annual growth rate of forest biomass stock reported in FRA was used as a substitute proxy for forest stock growth.

In cases where data on growing stock development were insufficient, inconsistent, or outdated, forest stock growth was set to zero. These cases are indicated by flag “Z” in the accompanying dataset.

The resulting values provide a harmonized estimate of country-level forest stock growth consistent with the conceptual requirements of the TiMBA model, namely the representation of biological forest growth independent of harvest-induced stock reductions. The values for forest growing stock and annual forest stock growth are provided in the accompanying table.

3. Flag System for Data Sources

The table entails the following flag system to denote the source and reliability of the data:

- **A:** Taken from FRA growing stock for the total forest area (million m³), using data for the reporting year 2020.
- **B:** Taken from FRA growing stock for the total forest area (million m³), but the reporting year is not 2020; it may be older (e.g., 2010).
- **C:** Taken from FRA growing stock for naturally regenerated and/or planted forest area (million m³).
- **D:** Calculated from the FRA biomass stock (t/ha) and converted to million m³.
- **G:** Calculated from Global Forest Watch 2000 based on living woody biomass density (t/ha) and converted to million m³.
- **M:** Stock growth capped at the midhinge multiplied by 2.5, where the midhinge is defined as (Q3–Q1)/2.
- **O:** Own data correction – if the area is equal to or greater than 1 and the growing forest stock smaller than 1, the stock is set to 1.
- **Z:** Stock growth set to zero due to insufficient or outdated data on forest growing stock development. This flag indicates that due to the lack of reliable data on stock growth, no estimation could be made.

References

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